JAMES ROLMER V. CATALAN

Full-Stack Developer & Systems Engineer | Manila, Philippines | jamescatalan19@gmail.com | github.com/jeymzaintsnitch

EXPERIENCE

Full-Stack Developer â **Independent & Client Work (2023 - Present)**

- Engineered fast, accessible client applications with React, TypeScript, and Tailwind CSS.
- Architected REST APIs, webhook consumers, and background job queues in Node.js.
- Modeled normalized PostgreSQL schemas with composite indexing and query profiling.

Frontend Lead â **SparkFest Hackathon Platform (2024 - 2025)**

- Built real-time schedule synchronization and rubric scoring views for hackathon judges.
- Established component standards, Git review workflows, and zero-downtime deploys.

Systems Developer â **Polytechnic University of the Philippines (2021 - 2025)**

- Created automated examination and rubric evaluation workflows for large student cohorts.
- Enforced transactional integrity on grading records using PostgreSQL row-level locks.

SELECTED PROJECTS

Vanguard â **Cloud Security Compliance Monitor (2025)**

- Multi-tenant cloud infrastructure compliance monitor evaluating resource configurations in real time.

PUP Assessment Portal â **Examination & Rubric Platform (2024)**

- Collegiate exam platform supporting 500+ concurrent student sessions with automated scoring.

PulseRoute â **Lightweight API Gateway & Reverse Proxy (2024)**

- High-throughput HTTP reverse proxy enforcing sliding-window rate quotas and health probing.

Hyperion Cache â **In-Memory Key-Value Store (2024)**

- Zero-dependency in-memory TCP store with O(1) hash table indexing and Doubly-Linked LRU eviction.

EDUCATION

Polytechnic University of the Philippines (PUP) â **BS in Information Technology (2021 - 2025)**